TAMU THINKTANK

TEAM PRESENTATIONS

WHO ARE WE?

Team Electra is a group of Texas A&M engineering students designing a compact, high-efficiency **bidirectional EV charger** for the 2026 IFEC competition. Our charger can both **charge a vehicle** and **send power back to the grid**, and our team focuses on practical hardware, control systems, and clean-energy design. We work together to build real technology that supports smarter and more flexible energy use.

INTRODUCTION & OVERVIEW

What's our challenge?

What’s a bidirectional charger?

A bidirectional charger is a device that can both charge an electric vehicle's battery and discharge power from it back to the grid, a building, or another device. Unlike standard EV chargers that only send electricity one way (to the vehicle), **bidirectional chargers enable "vehicle-to-grid" (V2G), "vehicle-to-home" (V2H), or "vehicle-to-load" (V2L) capabilities**. This allows EVs to function as mobile energy storage units that can help power your home during outages, reduce electricity costs by sending power back during peak demand times, or even supply energy to other equipment.

What’s our challenge?

Design a **cost-effective** bidirectional charger that **works globally (US & EU grids)**, emphasizing cost minimization while hitting high efficiency and density targets.

Real-world application: enabling V2G/V2H without breaking the bank (critical for widespread EV adoption.)

GOALS AND OBJECTIVES

OBJECTIVE 1

1kW bidirectional power capability

OBJECTIVE 2

Dual input: 120Vac (60Hz) & 230Vac (50Hz)

OBJECTIVE 3

Output: 400Vdc nominal (150-450V range)

GOALS AND OBJECTIVES

OBJECTIVE 4

Efficiency targets: >92% @ 1kW, >80% @ 100W

OBJECTIVE 5

Power density: 2 kW/L including cooling & EMI filters

OBJECTIVE 6

Galvanic isolation (Hipot test)

Overview - What's our solution?

GaN Totem-Pole PFC Front End

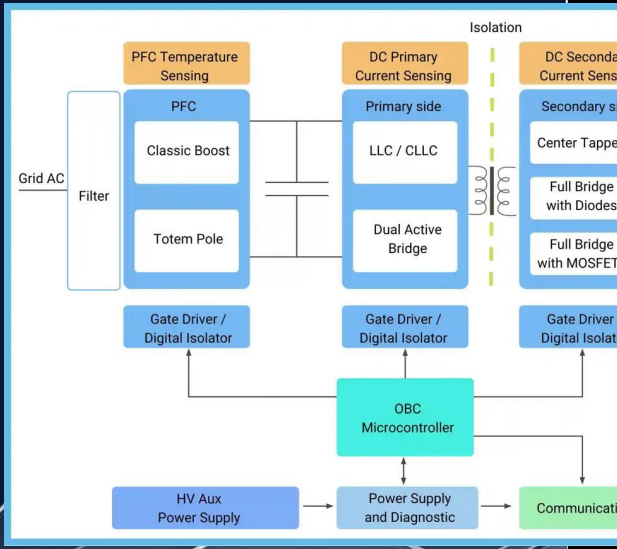
- Near-unity power factor, low THD, and no diode-bridge losses.
- Ultra-fast GS66508T GaN switches for high efficiency and low EMI.

Dual Active Bridge (DAB) Isolation Stage

- Soft-switching operation for high efficiency.
- High-frequency transformer for galvanic isolation and compact size.

Two-PCB Compact Mechanical Design

- Power stage on top PCB, control system on bottom PCB.
- Volume: 0.357 L ⟶ 2.80 kW/L power density.

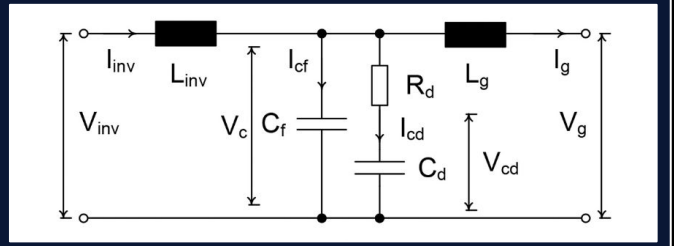


OUR SOLUTION

How does our charger work?

Stage I: AC Input/EMI Filter

The electricity in the grid is AC, but the battery of an EV needs DC in order to be charged. Before turning the AC into DC, the electricity needs to be as smooth as possible for efficiency. By using an electromagnetic interference filter (EMI Filter), it cleans the waveform of the electricity by getting rid of induced harmonics and unwanted frequencies disturbing the cleanliness. The design in our charger is an LCL filter with two shunt capacitors and a damping resistor. The LCL filter features bidirectionality as well as being cost effective and power efficient. The inductors are resistant to change in current, which means they are resistant to high frequencies. The lower frequencies pass through, and the shunt capacitors allows the high frequencies that got through to go through them and be dampened by the resistor. By using two shunt capacitors, it allows the capacitors to not be overloaded and lets the electricity be shunt evenly. The last inductor does one last filter, making sure most of the impeding frequencies are removed.



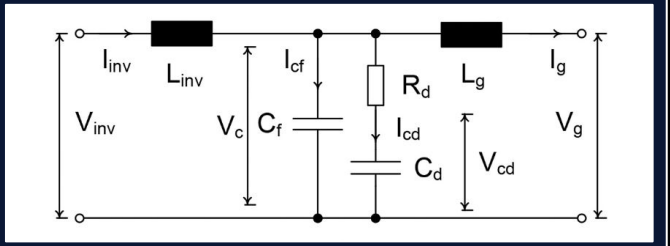
Stage 1: AC Input/EMI Filter

Why We Need an EMI Filter

- Grid electricity is AC, but EV batteries require DC.
- Before AC is converted to DC, the waveform must be as smooth as possible for efficiency.
  An EMI filter removes induced harmonics and unwanted high-frequency noise.

Our Filter Design: LCL EMI Filter

- Uses two shunt capacitors, two inductors, and a damping resistor.
- Provides bidirectional power flow, high efficiency, and lower cost.
- Works by:
  - Inductors resist changes in current → block high-frequency noise.
  - Lower-frequency components pass through the inductors normally.
  - Damping resistor gets rid of spikes in frequency or harmonics.
  - The capacitors block the low frequency and shunt the high frequency.
- Two shunt capacitors split the stress, preventing overload and balancing the filtering.
- Final inductor performs one last cleanup, ensuring most unwanted frequencies are removed.



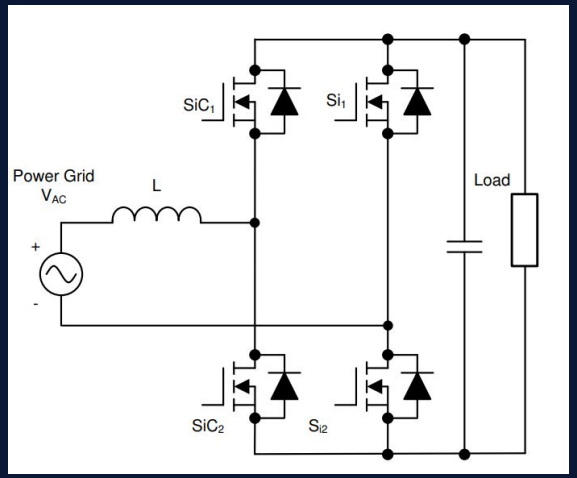
Stage 2: AC-DC/DC-AC

- Now that the incoming AC is smooth and filtered, it needs to be transformed into DC.
  - We need the voltage to be stepped up per contest specifications (120/230Vac → 400Vdc)

- Our design uses a totem pole power-factor correction circuit with gallium nitrate (GaN) MOSFETs.

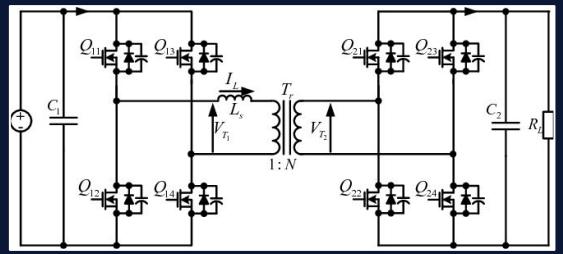
- How does the totem pole PFC circuit achieve our goal?
  - The boost inductor steps the AC voltage up.
  - The PFC, arranged in high and low frequency legs, employs high-frequency alternate switching to rectify the AC waveform into DC output.

- Why use a totem pole PFC?
  - The PFC structure with GaN MOSFETs allows for rectification in the charging direction and inversion in the discharging direction.
  - Conduction and reverse-recovery losses are minimized.
  - GaN MOSFETs allow for precise and active control over the PFC.



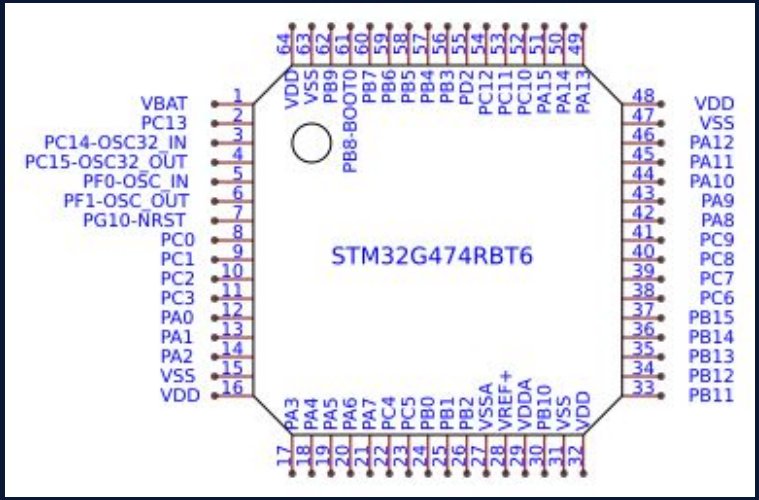
Stage 3: Dual Active Bridge

- What is a Dual Active Bridge?
  - An isolated DC-DC conversion stage for connecting a 400V DC bus (from PFC) to 300-450V battery pack with galvanic isolation for safety
  - Inherently bidirectional and can accommodate wide voltage charging range easily
- How does it work?
  - Phase-shift control (between 0 and 180 degrees) between the primary and secondary H-Bridge
  - Actually converts DC to AC then back to DC
    - Positive phase shift for grid to battery
    - Negative phase shift for battery to load
- Why use a DAB?
  - We can reach the galvanic isolation requirement with a smaller and cheaper transformer (does not need to convert 120V to 400V
  - Can achieve theoretical 98.2 % efficiency for the system)
    - Main losses stem for switching/ partial ZVS
  - Minimal thermal management is required



Microcontroller

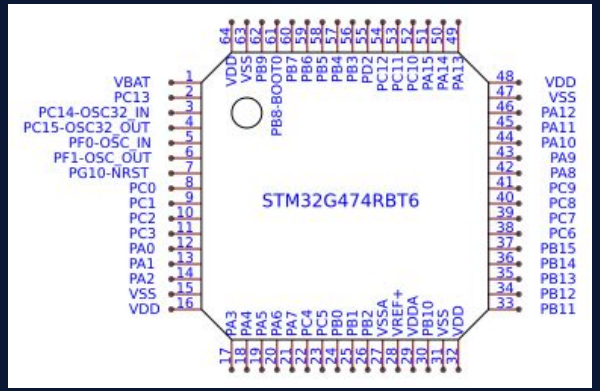
The microcontroller serves as the central controller of the bidirectional onboard charger, managing all real-time functions including high-resolution PWM generation for the Totem-Pole PFC and Dual Active Bridge stages. It processes AC and DC voltage, current, and temperature feedback through its high-speed ADCs, enabling accurate regulation and system supervision. The MCU also provides fast hardware-based protection using comparators and analog watch dogs, ensuring rapid fault response to overvoltage and overcurrent events. It oversees soft-start precharge sequencing, transitions between charge, discharge, standby, and fault modes, and maintains stable bidirectional operation. The STM32G474 microcontroller was selected for its strong real-time performance, strong analog interface, and advanced PWM features optimized for power electronics.



Microcontroller

- The central controller of the bidirectional onboard charger, manages all real-time functions including high-resolution PWM generation for the Totem-Pole PFC and Dual Active Bridge stages.
- Processes AC and DC voltage, current, and temperature feedback through its high-speed ADCs, enabling accurate regulation and system supervision.
- Provides fast hardware-based protection using comparators and analog watch dogs, ensuring rapid fault response to overvoltage and overcurrent events.
- Oversees
  - soft-start precharge sequencing
  - transitions between charge, discharge, standby, and fault modes
  - maintains stable bidirectional operation

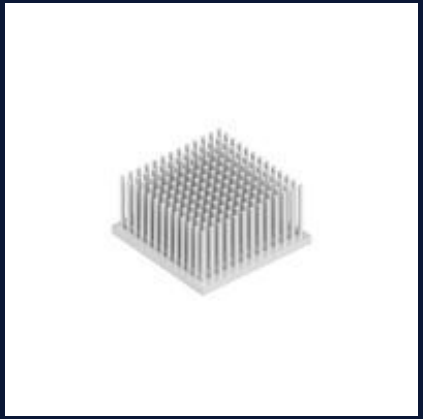
- The STM32G474 microcontroller was selected for its strong real-time performance, strong analog interface, and advanced PWM features optimized for power electronics.



BRIANNA LOPEZ

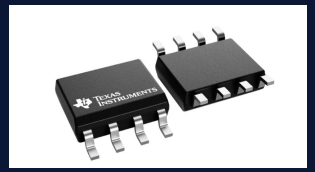
Heatsink

- The goal of a heatsink is to draw heat away from important components of a system in order to preserve their working capabilities.
- Under competition regulations, our design is only allowed to use convection to cool the charger.
- To keep our design easy to assemble and affordable, we are using aluminum plates with pins to maximize surface area for natural convection.
- Our design uses 3 heatsinks that are placed on top of each of the clusters of 4 MOSFETs. These are the main sources of heat and some of the most important pieces to keep under a safe temperature



Sensors

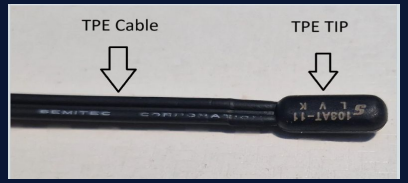
Our design uses three sensors: current, voltage, and temperature.



TMCS1100

Current Sensor — Hall-Effect

- Current creates a magnetic field that deflects electrons in the sensor.
- This deflection produces a measurable voltage, which is converted into current.
- Advantages:
  - Very accurate
  - Low power loss
  - Works bidirectionally for both AC and DC



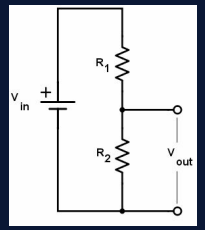
103AT-11

Temperature Sensor — Probe

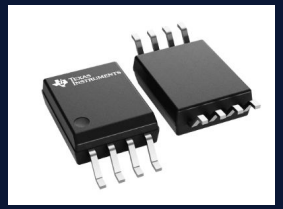
- Measures temperature near heat-critical components (MOSFETs, inductors, etc.).
  Why this design?
  - Fast response
  - Affordable
  - Reasonably accurate

Sensors

Our design uses three sensors: current, voltage, and temperature.



Resistor Divider



AMC1311

Voltage Sensor — Resistor Divider

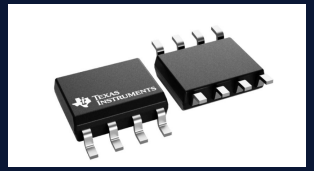
- A resistor divider scales a high voltage down to a lower, safe voltage.
- Allows the MCU to read voltage levels without being exposed to dangerous potentials.
- Why this design?
  - Extremely cheap
  - Very simple and reliable
    Easy to adjust by changing resistor values

Voltage Sensor — Isolated Amplifier

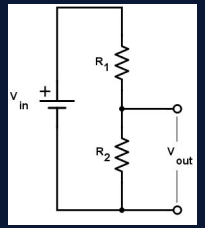
- Provides galvanic isolation between high-voltage and control circuitry.
  Ensures safe measurement and reduces noise coupling.
- Why this design?
  - Accurate
  - Durable
  - Protects the MCU from voltage spikes

Sensors

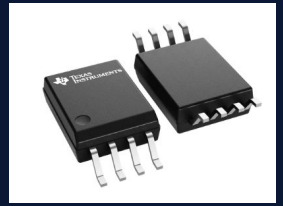
There are three different sensors used in our design: current, voltage, and temperature sensors. The design for the current sensor is a hall-effect sensor, where the current creates a magnetic field that deflects electrons to one side, which makes a measurable voltage. Then, the sensor converts the voltage into a value for current. This design is optimal because it has little power loss, is very accurate, and bidirectional for both AC and DC. The design for the voltage sensor uses a resistor divider and isolated amplifier. The resistor divider scales down the low voltage for the MCU to red, and the isolated amplifier galvanically isolates the voltage. This design is cheap, accurate, and durable. The temperature sensor is a probe that is quick, affordable, and fairly accurate.



TMCS1100



Resistor Divider



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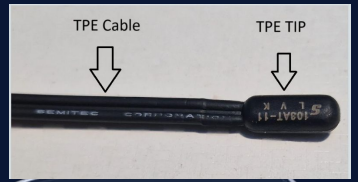

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SUBTITLE SLIDE TEMP

MINI DESCRIPTION OR NAME PRESENTING

ANNOUNCEMENTS

01

THINKTANK MERCH

02

RECURRING DESIGN CHALLENGES

03

MINI-DESIGN CHALLENGE COMPETITIONS

04

OFFICER POSITIONS FOR 2024-20225

05

PRESENTATION DIRECTIONS

WHAT ARE GOALS FOR

- Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis vulputate nulla at ante rhoncus, vel efficitur felis condimentum. Proin odio odio.
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PROJECT TIMELINE AND DELIVERABLES

How will we build our charger?

PCB Design (2 Layer PCB)

Top Layer: Power routing + components + high current paths

Bottom Layer: Ground plane + low-power control + sensing

Zone 1: AC-Input and EMI

- Fuse, relay, MOV, inrush circuit
- LCL EMI filter
- Located at board edge

Zone 2: Totem-Pole PFC

- GaN/SiC MOSFETs
- Boost inductor
- DC link capacitors

Design rule to keep note of: Keep switchin
cm² area

Zone 3: DAB Converter

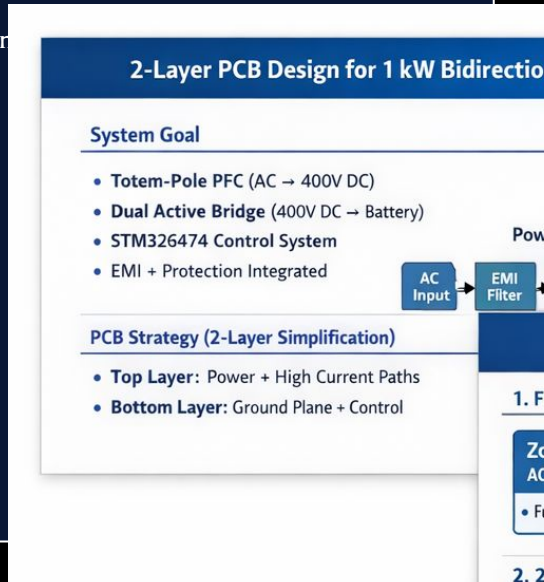
- STM32 MCU
- Gate drivers
- Sensors
- Isolated communication

MUST be separated from HV region

Zone 4: DAB Converter

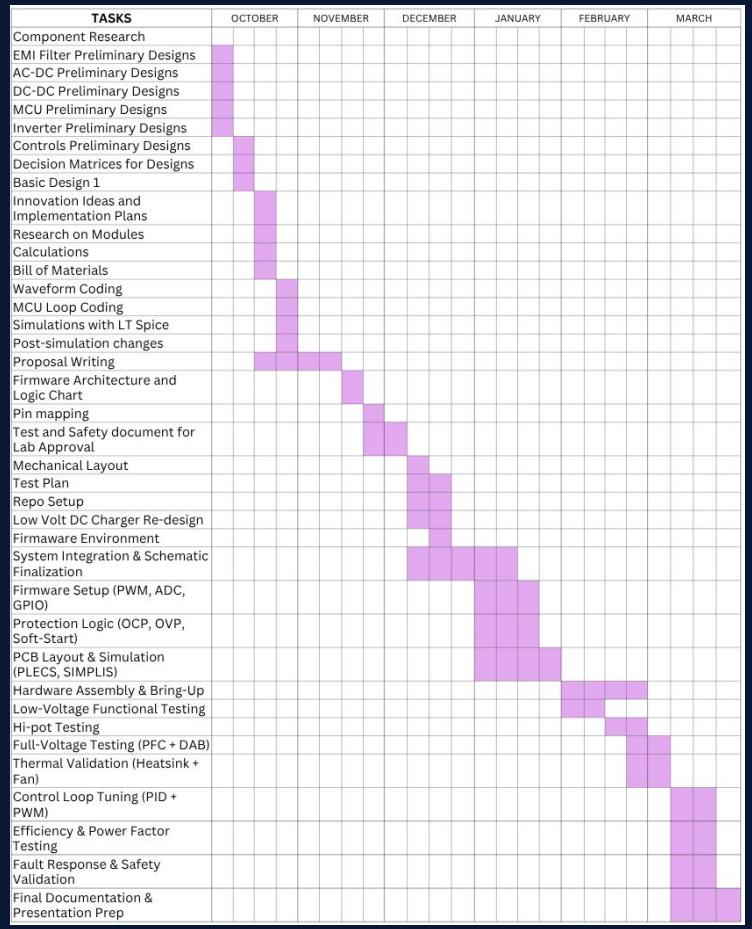
- STM32 MCU
- Gate drivers
- Sensors
- Isolated communication

MUST be separated from HV region



YOJITH MADDALA

Timeline


TASKS
OCTOBER
NOVEMBER
DECEMBER
JANUARY
FEBRUARY
MARCH
Component Research
EMI Filter Preliminary Designs
AC-DC Preliminary Designs
DC-DC Preliminary Designs
MCU Preliminary Designs
Inverter Preliminary Designs
Controls Preliminary Designs
Decision Matrices for Designs
Basic Design 1
Innovation Ideas and Implementation Plans
Research on Modules
Calculations
Bill of Materials
Waveform Coding
MCU Loop Coding
Simulations with LT Spice
Post-simulation changes
Proposal Writing
Firmware Architecture and Logic Chart
Pin mapping
Test and Safety document for Lab Approval
Mechanical Layout
Test Plan
Repo Setup
Low Volt DC Charger Re-design
Firmaware Environment
System Integration & Schematic Finalization
Firmware Setup (PWM, ADC, GPIO)
Protection Logic (OCP, OVP, Soft-Start)
PCB Layout & Simulation (PLECS, SIMPLIS)
Hardware Assembly & Bring-Up
Low-Voltage Functional Testing
Hi-pot Testing
Full-Voltage Testing (PFC + DAB)
Thermal Validation (Heatsink + Fan)
Control Loop Tuning (PID + PWM)
Efficiency & Power Factor Testing
Fault Response & Safety Validation
Final Documentation & Presentation Prep

Deliverables

	A	B	C
1	**Requirement**	**Status**	**Our Design**
2	Operation on 230 V AC (50 Hz) and 120 V AC (60 Hz) single-phase grid	Done	Universal input tested
3	Output 300–450 V DC during charging mode	Done	400 V DC regulated, tested
4	Galvanic isolation (≥ 1.5 kVAC hi-pot)	Done	1.5 kVAC high-frequency transformer isolation
5	Maximum output power of 1 kW	Done	1 kW continuous
6	Low-cost converter / manufacturability	Done	Cost-optimized PCB, BOM
7	High efficiency > 92 % peak	Done	96.05% (calculated)
8	High power density > 2 kW/L (including filters & cooling)	Done	2.8 kW/L (including cooling)
9	Soft start or pre-charge logic	Done	MCU-controlled soft start
10	Natural convection or self-supplied forced-air cooling	Done	Aluminum heatsink, forced air
11	Single 12 V DC auxiliary supply for electronics	Done	External 12 V
12	Overvoltage and overcurrent protection (OVP/OCP)	Done	Hardware & firmware based OVP/OCP
13	AC-side fusing (≤15A @ 230V)	Done	15A AC fuse
14	Connector compatibility (FCR series)	Done	FCR-compatible interfaces
15	Safety container / open-frame converter	Done	In accordance
16	No real battery used during testing	Done	Simulated battery
17	Bidirectional operation (V2L/V2G, 120 & 230 V)	Done	Bidirectional control & topology
18	High efficiency over load range (>92%@1kW, >80%@100W)	Done	96.05% @ 1kW, >80% @ 100W
19	Power factor correction (PF ≥ 0.96, THD < 5%)	Done	PF 0.99, THD < 5%
20	Electromagnetic compliance (EMC)	Done	EMI filter, pre-compliance results
21	Plug-and-play / automatic operation	Done	Automatic MCU operation
22	Integrated LVDC charger (12V or 48V @500W)	In Progress	Not yet implemented, redesigning

Bill of Materials

Item	Source	Quantity	Unit Price	Total Price (1000 Units)	Total Price
Totem-Pole PFC:					
Inductor	ITG Electronics	2	$15.00	$30,000.00	$30.00
Output Capacitor	Digikey	1	$12.74	$12,740.00	$12.74
C3M0065100K (Fast-Switching) MOSFETs	Digikey	2	$24.39	$48,780.00	$48.78
C3M0032120K (Slow-Switching) MOSFETs	Digikey	2	$22.21	$44,420.00	$44.42
DC Link:					
Input Capacitors	Mouser	2	$5.14	$10,280.00	$10.28
Output Capacitors	Mouser	2	$6.22	$12,440.00	$12.44
Bi-directional DC-DC:					
HF Transformer	Würth Elektronik (Custom)	1	$90.00	$90,000.00	$90.00
GS66508T-MR (Primary Side) GaNFETs	Mouser	4	$16.07	$64,280.00	$64.28
GS66508T-MR (Secondary Side) GaNFETs	Mouser	4	$16.07	$64,280.00	$64.28
Microcontroller:					
STM32G474RET6 MCU	Arrow	1	$4.32	$4,320.00	$4.32
Cooling:					
Heatsinks(Aluminum Tracks/blocks)	Digikey	3	$8.69	$26,070.00	$26.07
EMI Filter:					
Inductor Invertor Side	Mouser	1	$7.34	$7,340.00	$7.34
Inductor Grid Side	Mouser	1	$16.00	$16,000.00	$16.00
Capacitor	Digikey	2	$0.53	$1,060.00	$1.06
Resistor	Mouser	1	$3.43	$3,430.00	3.43
Sensors:					
Current Sensor	TI	2	$1.27	$2,530.00	$2.53
Resistor Top For DC Resistor Divider	Mouser	4	$0.14	$560.00	$0.56
Resistor Top For AC Resistor Divider	Mouser	4	$0.15	$600.00	$0.60
Resistor Bottom for both AC and DC Resistor Divider	Mouser	2	$3.50	$7,000.00	$7.00
Voltage IC Sensor	TI	2	$2.07	$4,140.00	$4.14
Temperature Sensors	Online Sensors	1	$6.00	$6,000.00	$6.00
Over Current and Voltage Protection:					
AC Relay	RS	1	$17.72	$17,720.00	$17.72
NPN to drive AC Relay	Digikey	1	$0.07	$70.00	$0.07
Resistor between NPN and Comparator	Mouser	1	$0.16	$160.00	$0.16
Comparator for both AC and DC	TI	2	$0.95	$1,900.00	$1.90
DC IC MOSFET driver	Digikey	1	$1.36	$1,360.00	$1.36
DC N-MOSFETS	Digikey	2	$12.14	$24,280.00	$24.28
Fuse before battery	Mouser	1	$2.34	$2,340.00	$2.34
Soft Start:					
Precharge Resistor	Digikey	5	$0.27	$1,350.00	$1.35
Precharge Relay	Digikey	1	$37.25	$37,250.00	$37.25
Capacitor	Mouser	2	$3.79	$7,580.00	$7.58
NPN to drive Relay	Digikey	1	$0.07	$70.00	$0.07
Resistor between NPN and Comparator	Mouser	1	$0.16	$160.00	$0.16
Miscellaneous:					
PCB	Lusher Engineering Services (Custom)	1	$10.00	$10,000.00	$10.00
Wiring (~20ft)	DigiKey	1	$11.99	$11,990.00	$11.99
AC Input Connector	Digikey	1	$7.01	$7,010.00	$7.01
Board-to-Board Connector	TestEquity	1	$0.36	$360.00	$0.36
Sensor Connectors	Mouser	3	$0.50	$1,500.00	$1.50
DC Input	Digikey	2	$0.22	$4,400.00	$0.44
			Totals:	**$585,770.00**	**$581.81**

EoS Presentation Q&A

GOT ANY QUESTIONS? WE'RE HAPPY TO ANSWER THEM!

FOLLOW OUR SOCIALS!

INSTA:

@tamuthinktank

WEBSITE:

https://www.tamuthinktank.org/

DISCORD:

QUESTIONS?

If you have any questions, feel free to reach out to us via Discord or email!

DISCORD:

EMAIL:

tamuthinktank@gmail.com